

Mohammed Hassan Mahmoud

AI Systems Engineer

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Professional Summary

Artificial Intelligence student specializing in building end-to-end AI systems across retrieval, machine learning, data engineering, and software development. Developed production-oriented platforms including a real-estate valuation system, an academic intelligence platform, and computer vision solutions using real-world datasets. Experienced in designing AI applications spanning data pipelines, intelligent retrieval, backend services, databases, and user-facing web and mobile applications.

Education

Bachelor of Science in Artificial Intelligence

Arab Academy for Science, Technology and Maritime Transport (AAST)

College of Artificial Intelligence – Intelligent Systems Track

Expected Graduation: 2026

Professional Training & Certifications

Professional Training

- ITI – Machine Learning (60 Hours), 2024
 - WE – Big Data Training Program, 2024
 - Microsoft × MCIT – Introduction to AI & Generative AI, 2025
 - E.STYRENICS – Information Systems Summer Training, 2025
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Projects

ValorAI

- Built an end-to-end AI-powered real-estate intelligence platform combining geospatial analytics, machine learning valuation models, explainable AI, and an AI copilot for property decision support.

- Developed a hybrid valuation architecture combining Comparable Market Techniques (CMT) with machine learning-based appraisal models for property valuation.
- Engineered geospatial intelligence workflows using PostGIS, H3 spatial indexing, CatBoost, and SHAP-based explainability.
- Designed and implemented an Agentic AI Copilot to assist users with property insights, valuation interpretation, and decision support.
- Built production-oriented backend services using FastAPI and developed user-facing applications with React and Flutter.
- Developed automated data acquisition, processing, and validation pipelines supporting a real-estate ecosystem built on more than 65,000 real property listings.
- Implemented intelligent routing mechanisms to dynamically select the most appropriate valuation pathway based on available property context.
- Platform is currently being evaluated and tested by real-estate professionals to improve valuation workflows and user experience.

AAST Academic Intelligence Platform (Graduation Project)

- Developed a full-stack academic intelligence platform combining GraphRAG, Knowledge Graphs, vector retrieval, recommendation systems, voice interfaces, and VR navigation to support student decision-making and academic guidance.
- Designed a hybrid AI architecture featuring Brain Routing, Orchestration, Intelligence Memory, Fusion Engines, Circuit Breakers, and hybrid retrieval pipelines across Neo4j, Qdrant, Ollama, and Gemini.
- Built a college recommendation and decision-support system that analyzes student interests, strengths, weaknesses, budget constraints, and career goals to recommend suitable colleges and academic tracks.
- Engineered automated data-ingestion, scraping, normalization, and knowledge-construction pipelines that transform raw university information into structured knowledge bases for retrieval and reasoning workflows.
- Developed voice-enabled interaction pipelines supporting speech-to-text academic advisory experiences and profile extraction workflows.
- Built a VR-based campus navigation subsystem enabling room discovery, interactive exploration, and guided navigation across university facilities.
- Integrated GraphRAG, semantic retrieval, vector search, and knowledge graph reasoning to improve answer grounding and reduce information ambiguity.
- Planned for integration within the College of Artificial Intelligence (AAST) to support future students and academic advisory workflows.

Petroleum Permeability Prediction System

- Delivered a paid machine learning project using real petroleum engineering data for permeability prediction and reservoir-related analysis.
- Performed extensive data cleaning, preprocessing, exploratory analysis, and feature engineering to improve weak feature-target relationships.
- Evaluated multiple machine learning approaches and neural network models to identify the most effective predictive solution.
- Improved model performance through iterative feature engineering, experimentation, validation, and optimization workflows.
- Built and deployed a prediction interface for practical usage and validation on unseen data.

F1 Vision Intelligence System

- Developed a computer vision pipeline for Formula 1 race analysis using YOLO-based detection, tracking, and classification workflows.
- Fine-tuned object detection models to improve recognition of Formula 1 vehicles and race-specific visual elements.
- Implemented team classification, damage detection, collision analysis, and overtake detection capabilities.
- Integrated tracking pipelines and telemetry estimation logic to analyze vehicle behavior across video sequences.
- Combined computer vision models with custom rule-based analytics to generate race-event insights from visual data.

Open Data Contributions

- Published large-scale real-estate datasets on Kaggle, including 20,000+ UAE property listings and 64,000+ Egypt property listings.
- Built automated scraping, cleaning, normalization, feature engineering, and PII-redaction pipelines used across both datasets.

Technical Skills

AI Systems

Agentic AI, RAG, GraphRAG, Knowledge Graphs, Hybrid Retrieval, LLM Applications, AI Orchestration

Machine Learning

Machine Learning, Deep Learning, NLP, Computer Vision

Backend & Frontend

FastAPI, React, REST APIs

Data & Databases

PostgreSQL, PostGIS, Neo4j, Qdrant, FAISS, Redis, SQL

Frameworks

PyTorch, TensorFlow, Scikit-Learn, Hugging Face, LangChain, Sentence Transformers

Tools

Docker, Git

Programming

Python, C/C++, JavaScript